Anay Gupta

469-486-2271 | anaygupta2004@gmail.com | linkedin.com/in/a-gupta123 | anaygupta2004.github.io

Education

Georgia Institute of Technology — Atlanta, GA
Bachelor of Science in Computer Science [Machine Learning Specialization]: 3.93/4.00 — *Aug 2022 – May 2025*

Skills

Languages: Python, SQL, Java
Developer Tools: Git, Docker, Google Cloud Platform (GCP), Amazon Web Services (AWS), Microsoft Azure
Recognitions: 2022 Barry Goldwater Scholar, 2023-2024 AACR Undergraduate Award, 2023 Canada Fulbright Mitacs Recipient

Experience

EntityML (YC F25) — San Francisco, CA
CEO, Co-Founder — *Jul 2025 – Apr 2026*

- Built a historical orderbook data API service cataloging 500K+ Polymarket/Kalshi markets while retaining 10 TB across 500M append-only S3 objects using Redis/WAL buffering, reconciliation, and gap recovery.
- Operated two generations of an authenticated market-data API serving hundreds of registered users, logging 5.86M requests with peak traffic of 845K requests/day and 99.3% uptime on the modernized service.

Amazon — Seattle, WA
Software Development Engineer Intern — *May 2025 – Aug 2025*

- Built a fullstack labor-planning configuration management system that streams changes through AWS Lambda into an append-only audit history table with reversion capabilities, processing thousands of updates/day across hundreds of stations worldwide in production.

BioML, Microsoft Research — Cambridge, MA
Research Intern — *May 2024 – Aug 2024*

- Developed new tokenization schemas for single-cell foundation models to try to leverage existing functional biological knowledge with new vocabulary for increased downstream performance, processing over 150B tokens
- Discovered that permutation-invariant single-cell foundation models with a masked objective reach SOTA performance with 80% fewer training steps [Adjacent Preprint #1, Adjacent Preprint #2]

Enterprise AI, Delta Air Lines — Atlanta, GA
AI Software Engineer Intern — *Jan 2024 – Apr 2024*

- Developed the first evaluation system for a RAG pipeline on AWS SageMaker, serving 20,000 customer service agents internationally

Biology Research AI Department (BRAID), Genentech — San Francisco, CA
Research Intern — *May 2023 – Oct 2023*

- Worked on a new framework Polygraph for evaluating synthetic DNA regulatory elements for cell-type-specific activity and "humanness" for gene therapy applications

Cancer Program, Broad Institute MIT and Harvard — Cambridge, MA
Research Intern — *Jun 2021 – May 2023*

- Completed single-cell bioinformatics analysis, examining 2M cells from 10 patients and 15 cancer cell lines pre- and post-treatment; performed differential expression, pathway enrichment, gene regulatory network inference, and cell-cell communication studies

Publications

Sycophancy Towards Researchers Drives Performative Misalignment. David D. Baek, Xinnuo Li, **Anay Gupta**, et al. *ICML 2026 Spotlight Poster*; arXiv:2606.08629.

Evaluating the role of pretraining dataset size and diversity on single-cell foundation model performance. Alan DenAdel, Madeline Hughes, Akshaya Thoutam, **Anay Gupta**, et al. *Nature Methods*, 2026.

Polygraph: a software framework for the systematic assessment of synthetic regulatory DNA elements. Avantika Lal, Laura Gunsalus, **Anay Gupta**, et al. *Genome Biology*, 2025.

Consequences of training data composition for deep learning models in single-cell biology. Ajay Nadig, Akshaya Thoutam, Madeline Hughes, **Anay Gupta**, et al. *bioRxiv*, 2025.

Drug persistence pathways in ovarian cancer identified by single-cell analysis of patient samples and cell line models. Elizaveta Leshchiner, Richard Panayiotou, **Anay Gupta**, et al. *Cancer Research*, 2023.